Title: Pneumonia Classification Using Convolutional Neural Networks

Module: Computer Vision

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Introduction

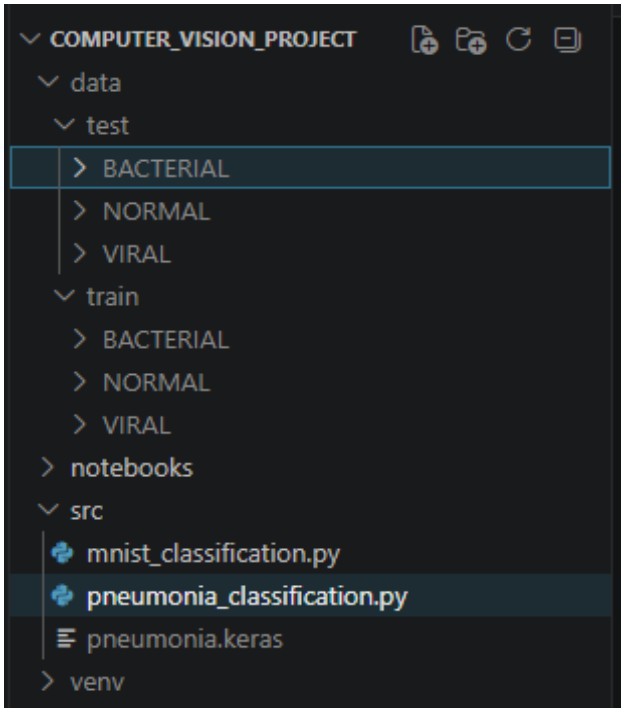
This project involves classifying pneumonia types (Bacterial, Viral, Normal) from chest X-ray images using Convolutional Neural Networks (CNNs) built with TensorFlow and Keras. The goal is to maximize classification accuracy and evaluate the model's performance using precision, recall, and F1-score.

Dataset

The dataset used consists of chest X-ray images classified into three categories:

- Bacterial Pneumonia
- Viral Pneumonia
- Normal

The data was split into training (80%) and validation (20%) sets. Test data was used for final model evaluation.



Model Architecture

The model is a Convolutional Neural Network (CNN) consisting of the following layers:

- Conv2D layers with ReLU activation for feature extraction.
- BatchNormalization layers to stabilize training and improve convergence.
- MaxPooling layers to reduce spatial dimensions.
- Flatten layer to convert 2D data into a 1D vector.
- Dense layer for the final classification.
- Dropout layer to prevent overfitting.

Model Architecture Summary:

- Input Shape: 128x128x3 (RGB images resized)
- 3 Conv2D layers
- 3 MaxPooling2D layers
- Dense fully connected layers
- Output: 3 classes (Bacterial, Viral, Normal)

Data Augmentation

Data augmentation was applied to artificially increase the size of the training dataset:

- Random horizontal flip

- Random rotation
- Random zoom

This helps the model generalize better and avoids overfitting.

Early Stopping

EarlyStopping was used to stop training if the validation loss didn't improve for 3 epochs. This prevents overfitting and helps in selecting the best model.

Training Process

The model was trained for 8 epochs using Adam optimizer and sparse categorical cross-entropy loss. Training accuracy and validation accuracy were monitored for overfitting.

- Epochs: 8
- Batch Size: 12
- Optimizer: Adam
- Loss Function: Sparse Categorical Cross-Entropy

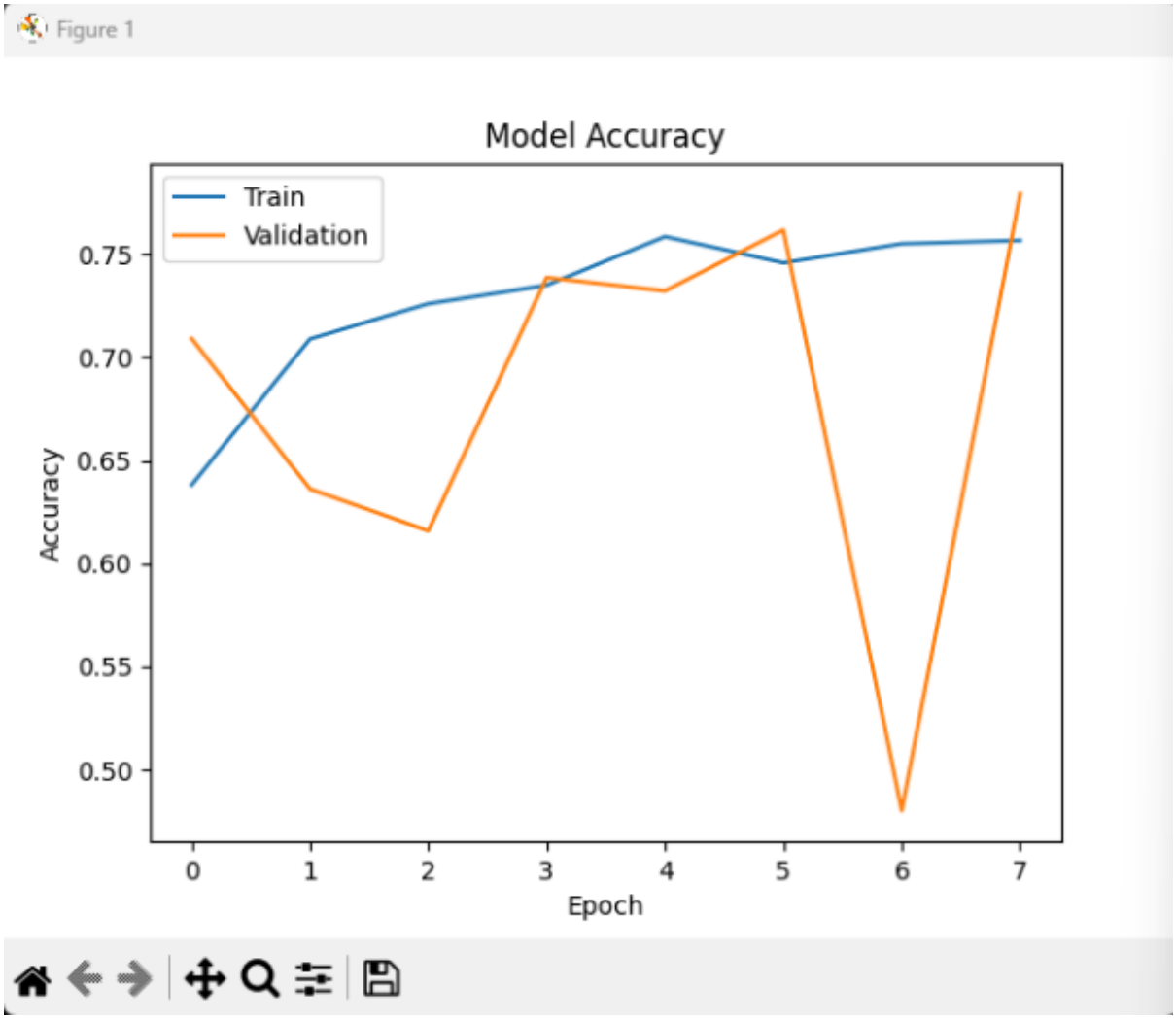


Figure 1: Training vs Validation Accuracy over Epochs

Results

The model achieved a test accuracy of 69.3%, which reflects its ability to correctly classify the pneumonia types.

Test Accuracy: 69.3%

Model Accuracy and Validation Accuracy were plotted to show training and validation performance.

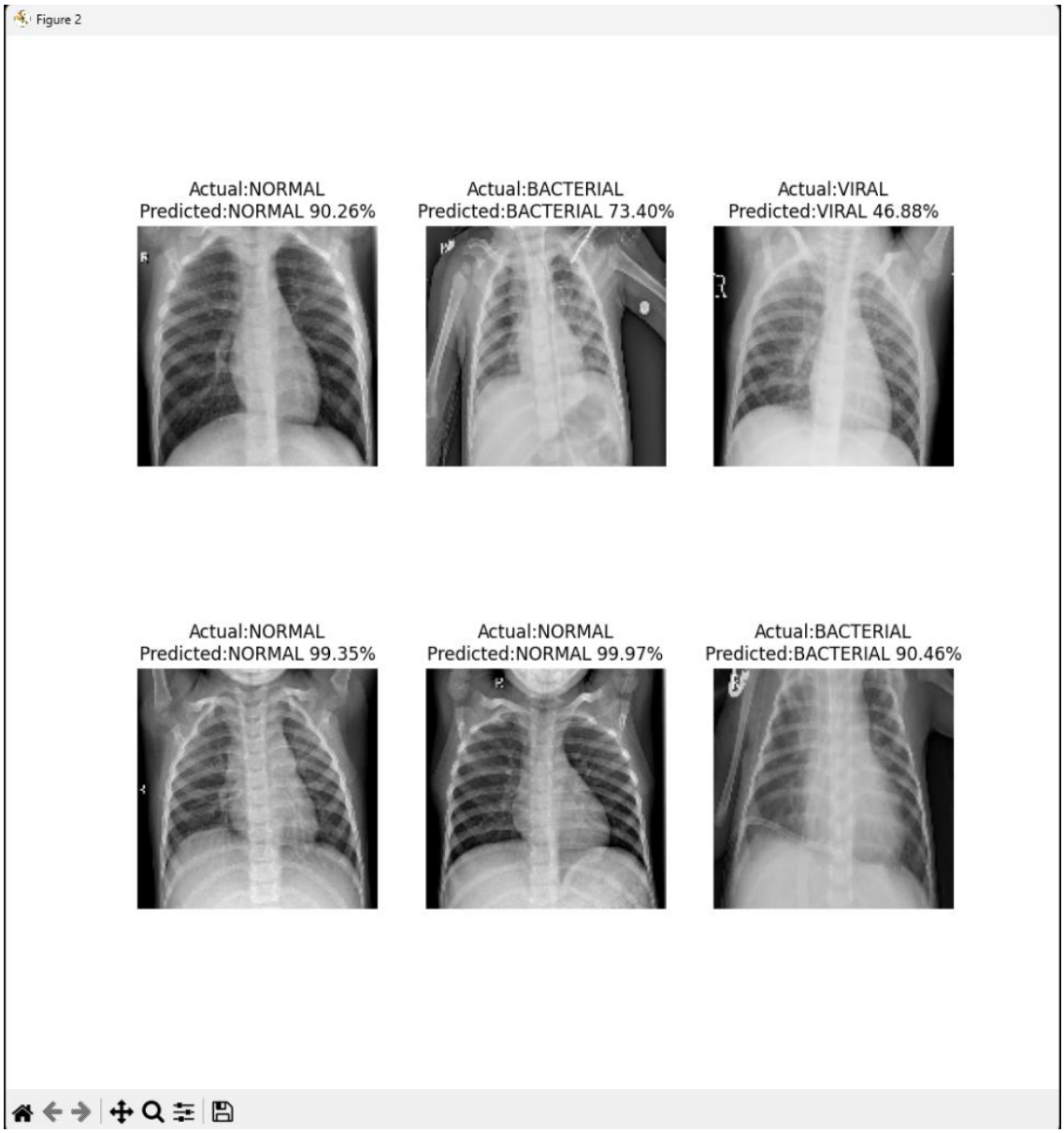


Figure 2: Sample Predictions Showing Actual vs Predicted Labels

Evaluation Metrics (Precision, Recall, F1-Score)

We calculated precision, recall, and F1-score for each class (Bacterial, Viral, Normal) using the classification report:

Precision, Recall, F1-Score for each class was printed as follows:

Class	Precision	Recall	F1-Score
Bacterial	**0.61**	**0.90**	**0.73**
Normal	**0.90**	**0.93**	**0.92**
Viral	**0.57**	**0.18**	**0.28**

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Classification Report:
              precision    recall  f1-score   support

   BACTERIAL       0.61      0.90      0.73       184
      NORMAL       0.90      0.93      0.92       122
       VIRAL       0.57      0.18      0.28       131

    accuracy                           0.69       437
   macro avg       0.70      0.67      0.64       437
weighted avg       0.68      0.69      0.65       437
```

Figure 3: Classification Report Showing Precision, Recall and F1-Score

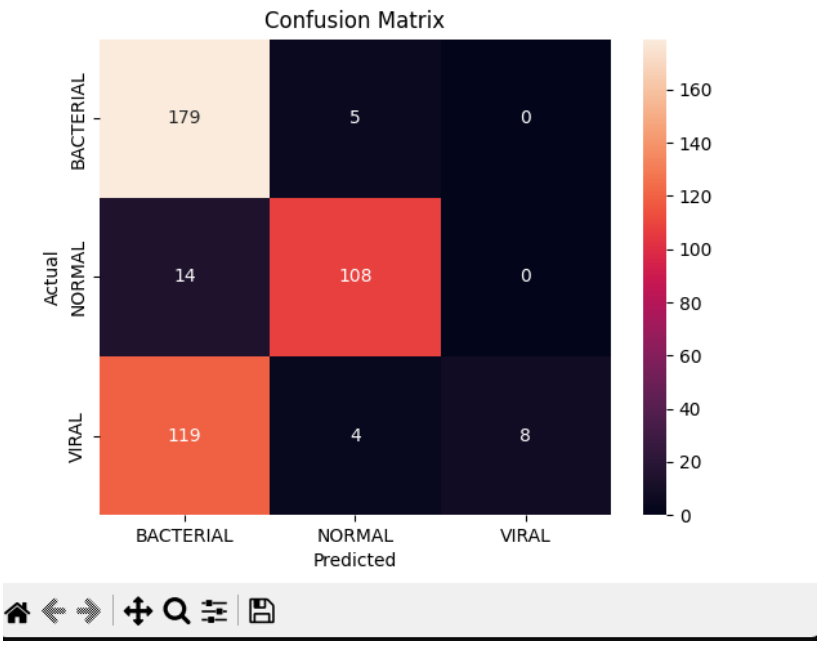

Figure 4: Confusion Matrix Showing Classification Performance Across Classe

Discussion

Overfitting:

Validation Accuracy fluctuated significantly, which suggests potential overfitting. This was addressed by using EarlyStopping and data augmentation.

Validation Instability:

A noticeable drop in validation accuracy is observed around epoch 6 in Figure 1. This indicates instability during training, where the model temporarily overfits to the training data and fails to generalize well to unseen validation data. This behaviour suggests that the model may be too sensitive to variations in the dataset. EarlyStopping was applied to mitigate this issue by preventing excessive training once validation performance stops improving.

Model Improvement:

The model's performance could be further improved by using transfer learning with a pre-trained model like ResNet or MobileNet, which can better generalize to unseen data.

Viral Class Performance:

The model performs poorly on the Viral class, with a low F1-score of 0.28. This is likely due to the visual similarity between Viral pneumonia and Normal chest X-rays, making it difficult for the model to distinguish between the two classes. In addition, the dataset is not perfectly balanced, and the Viral class has fewer samples compared to Bacterial cases. This may have limited the model's ability to learn strong distinguishing features for Viral pneumonia, resulting in low recall and overall performance for this class.

Conclusion

The CNN model was able to classify chest X-ray images into three categories with a final test accuracy of 69.3%. The model achieved strong performance on Normal and Bacterial classes, as reflected by their high precision and F1-scores. However, performance on the Viral class was significantly lower, indicating difficulty in distinguishing it from other classes.

The results highlight key limitations of the model, including class imbalance and the visual similarity between classes, particularly between Viral and Normal X-rays. In addition, instability observed in validation accuracy suggests that the model may be prone to overfitting.

Future improvements could include the use of transfer learning with pre-trained models such as ResNet or MobileNet, increasing the size and balance of the dataset, and applying more advanced regularization techniques. These improvements would likely enhance generalization and overall classification performance.